

Homework 1

Due: *Monday, May 24th at 11:59pm EDT*

All homeworks are due the Monday after their release at 11:59pm EDT on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit. We recommend checking out the sample proofs on the course website to see the level of rigor we're looking for.

Problem 1

- a. The Sun's favorite logical operator $\star$ has the following property on the logical propositions x and y

$$x \star y \equiv (\text{NOT } x) \text{ OR } (\text{NOT } y)$$

where $\equiv$ is the notation for equivalence, meaning the left and the right expressions are equivalent. Create a truth table for each of the following expressions, and state which single logical operator on x and/or y the expression is equivalent to.

- i. $x \star x$
 - ii. $(x \star y) \star (x \star y)$
 - iii. $(x \star x) \star (y \star y)$
 - iv. $(x \star (x \star y)) \star (y \star (y \star x))$
- b. The Sun has been awake for 4.603 billion years wondering the following: Can we express the OR operation using only IMPLIES operations? Justify your answer and help the poor Sun sleep.

Problem 2

For the first time in millions of years, the Sun, Mercury, Venus and the Earth are aligned. To celebrate this ‘syzygy’, they decide to play a dangerous game of asteroid gravitational deflection. The game has the following rules:

- I. Mercury loses if and only if the Earth gets hit with an asteroid.
- II. If the Earth orbits faster to get away, then Mercury deflects an asteroid towards Venus.
- III. If Mercury deflects an asteroid towards Venus, then the Sun yells at Mercury.
- IV. The Earth orbits faster to get away if and only if it does not get hit by another player with an asteroid.
- V. If the Sun yells at Mercury, then the Earth does not get hit with an asteroid.
- VI. Every player must either win or lose.
- VII. If Mercury does not deflect an asteroid towards Venus, then Venus wins.
- VIII. Only one player can win.

Using the eight rules, prove the following statements. Recall that if and only if statements require proof in both directions of implication.

- a. Mercury wins if and only if it deflects an asteroid towards Venus.
- b. The Earth gets hit with an asteroid if and only if the Sun does not yell at Mercury.
- c. The statement “The Earth does not win” is valid (under the definition given in lecture).

Problem 3

As a reminder, Conjunctive Normal Form (CNF) is a particular way of writing logical expressions. We define it as follows:

- A literal is a variable or its negation: x_1 and $\text{NOT}(x_1)$ are examples of literals.
- A clause is any disjunction (OR) of any number of literals: $(x_1 \text{ OR } x_2)$ or a lone literal are examples of clauses.
- An expression is in conjunctive normal form (CNF) if it is the conjunction (AND) of any number of clauses. The following is an example of a CNF expression.

$$(x_1 \text{ OR } x_2) \text{ AND } (x_3 \text{ OR } \text{NOT}(x_1)) \text{ AND } x_4$$

For each of the following expressions, convert it to CNF. Using truth tables or logical identities, prove your converted expression is logically equivalent to the given one.

- $(p \text{ AND } \text{NOT}(q)) \text{ OR } \text{NOT}(p)$
- $(p \text{ AND } \text{NOT}(q)) \text{ OR } (\text{NOT}(p) \text{ AND } q)$
- $\text{NOT}(p \text{ AND } (q \text{ OR } r))$
- $(p \text{ IMPLIES } q) \text{ IMPLIES } r$